

Rod Ciombor
2/15/26
CIS-2532-NET01
Dr. Sheikh Shamsuddin
Week 3 Lab

Question 1:

Write a program called SellItems. The program should allow the user to enter any item prices, calculate the total, and apply a 7% sales tax. The item prices should be stored in a Python list, and a loop should be used to get the input and calculate the total. The output should include a list of item prices, the subtotal of the prices, the tax amount, and the final cost. Display all results in two decimal places. Create at least two functions and save those functions into a separate file (the file that you created in your previous assignment). Import the file into your main program.

SellItems functions library contents

Here is a list of the contents in **SellItemsLib.py** which is imported into **SellItems.py**

```
In [ ]: # Author:      Rod Ciombor
# Date:        02/15/2026
# Instructor:  Dr. Sheikh Shamsuddin
# Class:       CIS-2532-NET01

SALES_TAX_RATE = 0.07

def getPricesFromUser():
    """
    This function prompts the user to enter an item cost.
    It then parses the value as a float and adds it to a list.

    It then asks the user if they would like to enter another item price,
    and the process will continue until the user
    enters something other than "Y" or "y"

    Upon completion, the function will return the list of item prices
    """
    items = []
    proceed = 'Y'
    #Prompt user for input
    while proceed == 'Y':
        itemCost = float(input('Please enter the price of an item: '))
        items.append(itemCost)

        print('Item price added to total')
        print()
        print('Would you like to enter another item price?')
        proceed = str(input('Enter "Y" for yes, "N" for no: ')).upper()
        print()
```

```

    print('Thank You!')
    print()
    return items

def displayItemPriceList(items):
    """
    This function displays a list of items, including the item number and price.
    """
    itemNum = 0
    for item in items:
        #print items of list
        itemNum += 1
        print(f'{"Item " + str(itemNum) + "":<15}${item:.2f}')
    print('-' * 30)

def displayTotalSection(items):
    """
    This function calculates the Sub-Total
    by adding up the values in a list of items.

    It then calculates the Sales Tax by multiplying
    the Sub Total by the Sales Tax rate.

    It then calculates the Total by adding the Sales Tax to the Sub-Total.

    It then displays this information formatted to two decimal places.
    """
    #Calculate and display Sub-Total
    subTotal = sum(items)
    print(f'{"Sub-Total:"<15}${subTotal:.2f}')

    # Calculate tax and add it

    salesTaxTotal = subTotal * SALES_TAX_RATE
    print(f'{"Sales Tax:"<15}${salesTaxTotal:.2f}')

    #Display Final Cost
    print()
    print(f'{"Total:"<15}${subTotal + salesTaxTotal:.2f}')

```

Main File Contents

Here are the contents of the main file, entitled **SellItems.py** with sample output

```

In [2]: # Author:      Rod Ciombor
        # Date:       02/15/2026
        # Instructor: Dr. Sheikh Shamsuddin
        # Class:      CIS-2532-NET01
        # Descr:
        #   Program prompts user to enter item prices and adds values to list
        #   It then calculates and displays the list of prices,
        #   the Sub-Total, the Sales Tax, and the final Total

import SellItemsLib as lib

```

```

def main():
    """
    This program prompts user to enter item prices and adds values to list.

    It then calculates and displays the list of prices,
    the Sub-Total, the Sales Tax, and the final Total.
    """

    #Prompt user for input
    items = lib.getPricesFromUser()

    #Print items of list
    lib.displayItemPriceList(items)

    #Calculate and display totals
    lib.displayTotalSection(items)

main()

```

Item price added to total

Would you like to enter another item price?

Item price added to total

Would you like to enter another item price?

Thank You!

Item 1: \$15.75

Item 2: \$30.00

Sub-Total: \$45.75

Sales Tax: \$3.20

Total: \$48.95

Question 2:

Write a program that prompts the user to enter two 3 x 3 matrices and displays their products.

MultiplyMatrix functions library contents

Here is a list of the contents in **MultiplyMatrixLib.py** which is imported into

MultiplyMatrix.py

```

In [ ]: # Author:           Rod Ciombor
        # Date:           02/15/2026
        # Instructor:   Dr. Sheikh Shamsuddin
        # Class:         CIS-2532-NET01

def validateMatrix(matrix):
    """
    This function checks if a matrix contains exactly 9 elements
    with numeric values.
    """

```

```

It will return False if any of these conditions are not met
'''

#Check if list is 9 items in length
if len(matrix) != 9:
    print('Matrix must contain 9 elements!')
    return False

for num in matrix:
    try:
        float(num)
    except ValueError:
        print('All values in matrix must be numeric!')
        return False
return True

def getMatrixInput(msg):
    '''
    This function prompts the user to enter a matrix of numbers,
    prompting the user to enter a new matrix if invalid data is entered.

    If the matrix is valid, it will parse all the elements as floats,
    pass the elements to a new list, then return that list

    The msg argument is the input message that will display
    to prompt the user.
    '''

    output = []
    isValid = False
    while not isValid:
        matrixInput = str(input(msg))
        matrix = matrixInput.split()
        isValid = validateMatrix(matrix)

    for num in matrix:
        output.append(float(num))

    return output

def reshapeMatrix(list, cols_per_row):
    '''This function reshapes a list into a 3 x 3 multi-dimensional list'''
    result = []
    for i in range(0, len(list), cols_per_row):
        upper = i + cols_per_row
        row = list[i:upper]
        result.append(row)
    return result

def multiplyMatrix(a, b):
    '''
    This function calculates the product of two algebraic matrices
    represented by multi-dimensional lists
    '''

    # Number of rows and columns
    rowsA = len(a)
    colsA = len(a[0])

```

```

colsB = len(b[0])

# Create result matrix filled
result = []
for i in range(rowsA):
    row = []
    for j in range(colsB):
        total = 0
        for k in range(colsA):
            total += a[i][k] * b[k][j]
        row.append(total)
    result.append(row)

return result

def displayResult(a, b):
    '''
    This function displays the product of two algebraic matrices
    represented by multi-dimensional lists
    '''
    print()
    print('The multiplication of the matrices is')
    product = multiplyMatrix(a, b)
    for i in range(len(a)):
        strA = " ".join(f"{x:.1f}" for x in a[i])
        strB = " ".join(f"{x:.1f}" for x in b[i])
        strProd = " ".join(f"{x:.1f}" for x in product[i])
        if i == 1:
            print(f'{strA} * {strB} = {strProd}')
        else:
            print(f'{strA}      {strB}      {strProd}')

```

Main File Contents

Here are the contents of the main file, entitled **MultiplyMatrix.py** with sample output

```

In [2]: # Author:      Rod Ciombor
# Date:      02/15/2026
# Instructor: Dr. Sheikh Shamsuddin
# Class:     CIS-2532-NET01
# Descr:
#   Program prompts user to enter two matrixes.
#   After validating the user input,
#   it reshapes each matrix into a 3 x 3 list
#   then multiplies them by each other and prints the result.

import MultiplyMatrixLib as lib

def main():
    '''
    This function prompts user to enter two matrixes.
    After validating the user input,
    it reshapes each matrix into a 3 x 3 list
    then multiplies them by each other and prints the result.
    '''

```

```
matrix1 = lib.getMatrixInput('Enter matrix 1: ')
matrix2 = lib.getMatrixInput('Enter matrix 2: ')

#Reshape both matrices to 3 x 3
matrix1 = lib.reshapeMatrix(matrix1, 3)
matrix2 = lib.reshapeMatrix(matrix2, 3)

#Multiply both matrices and display product
lib.displayResult(matrix1, matrix2)

main()
```

Matrix must contain 9 elements!

Matrix must contain 9 elements!

All values in matrix must be numeric!

The multiplication of the matrices is

1.0	2.0	3.0		0.0	2.0	4.0		5.3	23.9	24.0
4.0	5.0	6.0	*	1.0	4.5	2.2	=	11.6	56.3	58.2
7.0	8.0	9.0		1.1	4.3	5.2		17.9	88.7	92.4

In []: